Paidly — Product Architecture Strategy & Blueprint

Purpose: Shift from a *page list* mental model to **systems** and a defined **core engine**, while keeping the technical map honest.

PAIDLY ARCHITECTURE V2 (CLEAN REFERENCE)

*One-page upgrade summary. This section is the canonical **Paidly v2 system map** used across product and engineering.*

Paidly v2 Systems (final):

1. **Identity System** — Auth, users, organizations, roles, RLS-backed tenancy.
2. **Document Engine** — Shared compose, send, and PDF behaviour for commercial documents (invoices, quotes, payslips) **plus** a Documents Hub for other business document types. Persistence is split: specialised tables vs `documents`.
3. **Payment Intent Layer** — Canonical handoff from document delivery/observation to payment rails and settlement orchestration.
4. **Revenue System** — Payment providers, subscriptions, cash flow, and reporting (reads/rails downstream of documents).
5. **Relationship System** — Clients + catalog + offering intelligence that feeds document composition.
6. **Experience System** — Shared UI/interaction contracts (shell, sticky actions, money UX consistency).
7. **Payment Intelligence Layer** — Get Paid logic: reminders, nudges, triggers, and follow-up intelligence.

Frontend layout system (consistent everywhere):

- **Header** — title + primary **actions**
- **Content** — tables, grids, main **forms**
- **Side panel** — **summary**, filters, secondary controls (`PageTemplate` in code)

Data flow (client):

```
UI → Hooks → Services → Entity facades (EntityManager) → Supabase
```

**Serverless APIs (`/api/`)* handle:

- Payments (**Payfast**)
- Emails (**Resend** / SMTP)
- Public document access (shares, tokens)
- **Cron** jobs (dunning, reminders, ops)

Deployment note: Same-origin `/api` on Vercel is implemented by ** `api/*.js` *serverless functions**; **the optional** *Express **app in** `server/src/index.js` **is a** separate **runtime (rate limits and middleware apply only where that process fronts** `/api`**). See** `docs/API_DEPLOYMENT_MODEL.md`* before tuning limits or tracing 429s.

Goal: Evolve Paidly from an **invoicing tool** into a full **business operating system**—same story as **Positioning** below.

Final surface architecture (canonical)

Paidly is one product with **two workplaces** on **one core**. The till is not a second app and not a second database.

```
PAIDLY
│
├── BACK OFFICE          (sidebar / Settings — issue, stock, AR, people)
│   ├── Products / Catalog
│   ├── Inventory
│   ├── Customers
```

```
│   ├── Suppliers
│   ├── Purchase Orders
│   ├── Budgets
│   ├── Invoices
│   ├── Quotes
│   ├── Payslips
│   ├── Reports
│   └── Settings
│
└── RETAIL FRONT OFFICE  (dedicated `/POS` shell — no sidebar)
    └── POS
        ├── Product Search
        ├── Barcode
        ├── Cart
        ├── Customer
        ├── Discounts
        ├── Cash
        ├── Payment Intent
        ├── Payment Provider
        ├── Receipt
        ├── Returns
        └── Register

SHARED CORE (not duplicated per workplace)
├── Authentication
├── Organizations
├── Companies / Brands
├── Products            → public.services
├── Inventory           → services.stock_quantity + inventory_movements
├── Customers           → public.clients
├── Payment Intents     → payment_intents
├── Payments / Settlement  (invoice payments vs till pos_sales_events vs SaaS payment_history)
└── Reporting
```

Nav copy: **Products** (catalog page `Services`), **Clients** (customers). Staff open the till from back-office **POS**; they do not run invoices from till chrome.

Payment architecture — two completely separate domains:

```
Paidly Subscription
  → PayFast
  → Paidly subscription billing     (subscriptions + payment_history)

Customer POS Payment
  → payment_intent
  → POS payment rail                (cash | ozow | card_terminal)
  → provider
  → verified settlement
  → POS sale completed              (pos_sales_events)
```

Do not route till money through PayFast. Do not write till sales into invoice `payments` or SaaS `payment_history`. Invoice customer capture (when used) is also not PayFast on `payment_intents` — Ozow for documents; PayFast stays SaaS.

On-call session/runtime guardrails: see `docs/SESSION_RUNTIME_GUARDS.md`.

New to the codebase: see **`docs/HOW_EVERYTHING_CONNECTS.md`** (one-page diagrams: browser ↔ Supabase ↔ `/api` ↔ optional Express).

Runtime scalability & coordination: `docs/runtime-audit-report.md`, `src/core/` (`RuntimeCoordinator`, `RealtimeManager` logical registry over `paidlyRealtimeManager`, query policies, mutation dedupe, request budgeting, error classification).

Marketing SEO / structured data: homepage JSON-LD (`Organization`, `WebSite`, `WebApplication` offers) follows Google structured data policies — visible content only, no fabricated ratings. See `docs/SEO_STRUCTURED_DATA.md`.

POSITIONING — REAL PRODUCT ADVANTAGE

Paidly is not an invoicing app. It is **Business Management + POS** — not a POS product that happens to invoice. One organization can serve:

Business	Paidly
Consultant / freelancer	Invoices + quotes
Service business	Invoices + clients + expenses (no till; `business_type = service`)
Retail	POS + inventory + customers + payments (`retail`)
Hybrid	POS + invoices + quotes + inventory (`mixed`)

The hybrid path is the interesting one: a salon, clothing store, or agency can take **walk-in till sales and normal invoicing** on the same account. The platform loop is **Catalog → POS → Payment Intent → Settlement → Inventory → Reporting**. **PayFast stays isolated** as Paidly's subscription billing rail.

Investor-facing framing: shift the story from "invoice app" to **financial workflow platform for SMEs**.

You already have (in shipped or advanced form): invoices, quotes, payslips (specialised commercial tables), Documents Hub (generic business documents), clients, catalog/inventory, **native POS till** (plus Yoco/Square adapters), reporting, and the plumbing for payments, subscriptions, and cash visibility.

Competitive edge: most tools **excel at one slice** (invoicing-only, or accounting-only, or a disconnected referral program). **Few competitors unify** issuance (document engine), relationship/catalog input, revenue/ops read models, and payment intelligence **under one coherent architecture** and vocabulary. That unification—**Document → Payment Intent → Revenue System** plus shared **Experience** and **Payment Intelligence**—is the defensible story: not "another invoice PDF," but **operating the business** in one system.

Execution reality: roughly 70% of the SaaS architecture is already in place. The remaining value-defining 30% is: payment abstraction, event tracking, and experience consistency.

EXECUTIVE FRAMING

Old lens	New lens
"Invoicing app" / feature checklist	**Business operating system** — documents + relationships + money + growth
"We have Invoices, Quotes, Payslips…"	**One Document Engine** (shared compose/send/PDF) with **split persistence**
"Invoices page, Quotes page, Clients page…"	Commercial lists over specialised tables; Documents Hub for other business documents
Features as separate products	**Five core systems** + one **Experience system** (cross-cutting)
Routes drive architecture	**Capabilities** drive architecture; routes are just entry points

Brutal truth the strategy fixes: the stack was described accurately but not **product-driven**: no named "engine," no explicit system boundaries, and no place for **scaled UX consistency** beyond ad hoc pages.

Part A — Product architecture (systems)

A.1 PAIDLY V2 SYSTEMS (FINAL ARCHITECTURE)

These are the **real** architecture—not the sidebar.

1. Identity System

Job: Who is acting, for which organisation, with what authority.

- Supabase Auth (sessions, JWT)
- `profiles`, `organizations`, `memberships`, roles (`admin`, `management`, …)
- RLS as the enforcement layer; client as the UX layer

Organisation vs company/brand vs team (do not confuse these):

Concept	Meaning	Persistence
Organization / account	The Paidly tenant using the product	`organizations`; `CompanyContext.companyId` is this **org id** (product copy often says `company_id` for the tenant). `business_type` (`service` \
Company / brand	A trading identity that belongs to that organization, used for document branding **and POS till identity**	`public.companies` (`org_id`, `name`, `logo_url`); invoices set **`invoices.company_id` → `companies.id`**; POS registers and optional private catalog rows use the same id (`pos_registers.company_id`, `services.company_id`)
Team / membership	People who have access to the organization	`memberships` + org roles

Do not create a second tenant system. `CompanyContext` remains org RBAC. Active brand is a **UI default for new documents** (per-org localStorage); changing it must not rewrite existing invoices **and must not change which products a POS till can sell**. Quotes have no `company_id` column (name/logo snapshot only). Payslips stay organization-profile scoped. See `docs/MULTIBRAND_COMPANIES.md`.

Strategy: Stabilise, don't expand. Harden session edge cases, org bootstrap, and role gates (`RequireAuth`) so every other system trusts Identity cheaply.

Auth + session stabilization (in flight): one coherent story for **session read/write** (`getStableSession` / SessionCoordinator vs reserved raw `getSession` in AuthContext / SupabaseAuthService / supabaseAuthRefresh), **profile restore** when local auth cache is cleared, **invite / password / org bootstrap** flows, and **no silent "logged in but empty user"** states. Client work touches **`customClient.js`** / `AuthManager`, **`RequireAuth`**, and Supabase Auth config—treat this as **foundation** before shipping net-new document features at scale.

Connection lifecycle (client OS): `ConnectionLifecycleManager` (`src/lib/connection/ConnectionLifecycleManager.js`) is the single ingress for **auth, refresh, realtime, visibility, network, sleep/wake, and recovery** signals; it updates a read model store and delegates terminal decisions to `SessionOrchestrator` only through one adapter (`registerSessionAuthority(...toSessionAuthorityAdapter())`). Subsystems **report in**; they do not apply parallel session health policies. Transport events use semantic types (`LifecycleSignalType`, e.g. `REALTIME_DISCONNECTED`, `REFRESH_SKIPPED`) and **`lifecyclePolicy`** chooses effects (ignore vs recover vs reconnect) so realtime/visibility noise does not escalate to logout. The same signals also feed **`RuntimeCoordinator`** (`runtimeCoordinatorBridge.js`) for **phase + `pauseNonCriticalRequests`** so authenticated HTTP (`apiRequest` / `backendApi`) waits during AUTH_RECOVERING / RECONNECTING. **Session health** uses a degraded ladder (`sessionRecoveryEscalation.js` + `SESSION_STATUS`: `unstable` → `reconnecting` → `degraded` → `reauth_required`) before `transitionToExpired` for recoverable failures (reconnect, wake, reauth-class 401s); fatal refresh / explicit sign-out still expire immediately.

Sleep/wake app recovery lock: **`AppRecoveryLock`** (`src/lib/recovery/appRecoveryLock.js`) + **`wakeRecoveryStore`** enforce phased recovery: **`wakeRecoveryGuard`** blocks **`customClient`** mutations, **`SyncEngine`** skips queue draining while `blockMutations` is set, **`paidlyRealtimeManager`** suppresses postgres_change delivery during the lock, and **`runWakeRecoverySequence`** restores the JWT/session first (`lockPhase: auth`), then **`awaitRealtimeRecoveryHandlers`** + **`waitForPaidlyMainChannelJoined`** (`lockPhase: realtime`) before unlocking the UI.

Realtime + JWT rotation: On every successful refresh (`TOKEN_REFRESHED`, **`refreshSession`** success, optional **`USER_UPDATED`**), call **`reconcilePaidlyRealtimeAfterTokenRefresh`** (`paidlyRealtimeManager.js`): **`supabase.realtime.setAuth(access_token)`**, then **fully remove** the multiplex channel (**`removeChannel`**) and **recreate** it so postgres_changes always bind to the fresh JWT (no "joined but stale auth" shortcuts). Rebuild debouncing, heartbeat stale detection, transport error backoff, and a small **`RealtimeConnectionPhase`** state machine (**IDLE /**

CONNECTING / CONNECTED / STALE / RECONNECTING / FAILED) live in `paidlyRealtimeManager` + `paidlyRealtimeConnectionMachine.js`; `ConnectionMonitor` maps UX only (no parallel realtime reconnect timers). Visibility alone does **not** run a dedicated "stale realtime repair" path — recovery is heartbeat / transport / auth–driven. Use `validatePaidlyRealtime()` for a cheap health snapshot.

Background session resync: `sessionRefreshScheduler` (`src/lib/session/sessionRefreshScheduler.js`) is the single coalescing entry for **refresh + profile + realtime hint** work driven by visibility, online, heartbeat, keep-alive, tab sync, wake follow-ups, etc.; initiators call `requestSessionRefresh({ source })` instead of invoking `refreshSession` directly (the executor is registered from `AuthContext.impl.jsx`).

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2. Document Engine (the commerce core)

Job: Everything a business **issues to another party** to record obligation, intent, or compensation—then **delivers**, **tracks**, and often **collects** on.

Canonical product vocabulary (feature → source of truth):

Feature	Persistence (system of record)
Invoice	`invoices`
Quote	`quotes`
Payslip	`payslips`
Recurring invoice	`recurring_invoices`
Other business documents (leave, expenses, contracts, …)	`documents` (+ `document_items`, `document_events`)

Do not dual-write. Invoices, quotes, payslips, and recurring invoices must not be copied into `documents`. The Documents Hub is not a second store for commercial documents.

Persistence: commercial documents remain on specialised Supabase tables. Shared UI and helpers live in `src/document-engine/`. Ownership policy: `src/document-engine/documentSystemOfRecord.js`.

Migration status (implementation):

- **Shared now (code-level contracts, split persistence):**
- Shared document type vocabulary and routing helpers in `src/document-engine/`.
- Shared list-controller and adapter pattern for invoice/quote/payslip list screens.
- Shared pagination and shared export assembly service paths.
- **Documents Hub (generic types only):**
- Leave requests, expense claims, contracts, and other catalog types persist in `public.documents`.
- Hub **Convert to invoice / quote** (job cards, reports, proposals, scopes) opens specialised compose (`CreateDocument/invoice|quote?fromHubDocument=`) and writes to `invoices` / `quotes`. It does not insert commercial types into `documents`.
- Quote → invoice remains on the Quotes page (`CreateInvoice?quoteId=` / `CreateDocument/invoice?quoteId=`).
- Leftover `documents` rows with `type=invoice|quote|payslip` (if any exist from earlier experiments) are **hidden from the hub list**. Opening a direct URL shows a cleanup screen: go to the specialised list, or archive/remove the leftover hub row. Rows are not migrated and not auto-deleted.
- **Not a goal:**
- Migrating invoices, quotes, or payslips into `documents`.
- Dual-write / automatic mirroring of commercial documents into the hub.

Reframe: one **engine** (compose, send, PDF) with **two persistence owners**:

Document kind	Today's surface	Same engine primitives
Invoice	Invoices, recurring	Draft → send → view/track → pay / remind
Quote	Quotes, templates	Draft → send → accept/expire → convert

Document kind	Today's surface	Same engine primitives
Payslip	Payslips	Draft → send → employee view

Shared lifecycle (conceptual):

`Author` → `Compose` (lines, tax, branding) → `Render` (PDF/HTML) → `Deliver` (email, link, portal) → `Observe` (opens, reminders, delivery/engagement telemetry) → `Payment Intent` (amount/currency/expiry + rail handoff) → `Settle` (payment, acceptance, archive)

Observe layer (formalized)

`Observe` is not a loose log. It is a first-class event layer that records delivery, engagement, and money-adjacent milestones in a single stream per document.

Schema upgrade (`document_events`):

- `id`
- `document_id`
- `event_type` (`sent` | `opened` | `clicked` | `paid` | `reminded`)
- `occurred_at`
- `actor_type` (system | recipient | user | webhook)
- `metadata` (jsonb for channel, provider payload refs, reminder run id, etc.)

Event taxonomy (minimum canonical set):

- `sent`
- `opened`
- `clicked`
- `paid`
- `reminded`

This powers:

- Timeline (client and document history)
- Notifications (state-aware in-app/email nudges)
- Smart reminders (behavior + due-date + payment-intent aware)
- Analytics (delivery-to-payment funnel and conversion diagnostics)

Why this matters for product:

- One **mental model** for PM, design, and eng.
- One place to invest: **send pipeline**, **PDF pipeline**, **public token model**, **status vocabulary**, **line-item model**.
- One observable event stream (`document_events`) that unifies communication and payment lifecycle telemetry.
- Feature parity (e.g. quote send = invoice send) becomes **engine work**, not three copies.

Technical anchor today: `Invoice` / `Quote` / `Payslip` entities + `InvoiceSendService`-style orchestration + `/api/send-email` + public share routes. **In code:** `src/document-engine/` exports `DOCUMENT_TYPES`, `normalizeDocumentType`, `parseRouteDocumentTypeStrict`, `getDocumentEntity`, `documentRef`—use these for routing, analytics, and new engine code. **Roadmap:** grow this module (shared send/PDF adapters, shared status vocabulary) so new document kinds plug in, not fork.

Critical addition: Payment Intent layer (Document Engine ↔ Revenue & Ops)

Without a first-class `Payment Intent`, payments feel bolted on, Payfast-specific logic leaks across product surfaces, and the engine is harder to scale. Introduce a canonical handoff object between document delivery/observation (and native POS) and financial capture.

Why this matters:

- **Without it:** payments feel bolted on, provider logic leaks everywhere, and cross-surface consistency breaks.

- **With it:** clean abstraction between document/POS and payment rails, multi-provider readiness, and analytics-ready payment funneling.

Payment intent contract:

- `intent_id` (idempotent)
- `source_kind` (`document` | `pos`)
- `document_ref` (`type`, `id`, `org_id`) when `source_kind = document`
- `amount_snapshot` + `currency_snapshot`
- `payer_context` (public share, portal user, signed-in cashier, or walk-in)
- `provider` / rail: **customer payments** use `cash` (till-verified), `ozow` (Digital Payment), or `card_terminal` (physical reader — not click-to-paid). **PayFast is only for Paidly platform subscriptions** — never POS or invoice customer capture.
- `expires_at`
- `status` (`pending`, `requires_action`, `processing`, `paid`, `failed`, `cancelled`, `expired`, `refunded`)

POS payment path (Cash | Card | Digital Payment):

The native till Pay sheet offers three architecture-ready methods. **Cash** is the only till-verified settlement: the cashier counts notes, POS computes change, and the intent is `paid` after that till check. **Card** and **Digital Payment** are not "tap to mark paid." They stay unpaid until a real mechanism confirms.

Till method	`payment_intents.provider`	Completes sale when	Must not
Cash	`cash`	Cashier tenders ≥ total (trusted till cash workflow)	Route through Ozow or PayFast
Card	`card_terminal`	Hardware reader / webhook confirms (`terminal_confirmed`)	Cashier click, `manual_complete`, `force_paid`, or `mark_paid`
Digital Payment	`ozow`	Ozow charge/webhook confirms success	Complete from a till click alone

There is **no** `MANAGE_POS_MANUAL_CARD` (or similar) permission. Paidly does not fake a card terminal. External Yoco/Square readers are adapters: they confirm completed sales through `/api/pos/webhook`, which is a real confirmation — not the native **Card** button. Do not write till money into invoice `payments` or SaaS `payment_history`.

POS inventory commit:

Stock lives on `services.stock_quantity` and the ledger `inventory_movements`. POS uses the existing RPCs `adjust_inventory_stock` → `apply_inventory_movement` (`source = pos`, `reference_id = pos_sales_events.id`). Inventory **must not** decrease because a product was added to the till cart, Pay was opened, or a `payment_intents` row was created. Those steps are local/pending only.

```
Cart (device state)
  → Payment (intent pending)
  → Verified payment / confirmed cash
  → Sale completed (`pos_sales_events`)
  → Inventory movement
  → Stock decreases
```

Returns restock (`type = in`) only when the original sale already applied inventory. A return is a **new** `pos_sales_events` row (`sale_kind = return`, `parent_event_id` → original). The original sale is **never deleted** and its money columns are immutable (trigger). A failed inventory commit reverses any partial `out` so the catalog is not left half-moved. Failed return events do not consume remaining qty.

POS receipts: After a completed `pos_sales_events` sale (or return), the till issues a **customer receipt** — brand, sale number, date/time, staff, products/qty/prices, discount, tax, total, payment method, and change when cash. Staff can print, download PDF (same html2pdf / Anvil pipeline as other documents, without invoice templates), or email via `POST /api/pos/receipt/email` (Resend HTML + optional PDF). A receipt is not an invoice: checkout does **not** insert `invoices`, does not use `/api/send-invoice`, and does not treat a receipt as AR.

POS vs invoice (hard rule): A POS sale is an immediate retail transaction (`pos_sales_events`). An invoice is a receivable / payment request (`invoices` + `payments`). Do **not** auto-create an invoice for every till sale. After a completed sale, staff may optionally **Customer requests invoice** (`POST /api/pos/invoice`) if a named `clients` row is attached (walk-in cannot convert). That inserts a **paid tax-invoice copy** linked by `invoices.pos_sale_event_id` (unique) and `pos_sales_events.invoice_id` . It does not write invoice `payments` , does not send a Pay-now request, and does not move stock again (`handle_invoice_paid` / `handle_invoice_reversal` skip POS-origin rows). Cash flow ignores those invoices so till money is not counted as invoice income. Quote “Convert to Invoice” (unpaid compose from `?quoteId=`) is the wrong path for a settled till sale — that would ask the customer to pay again. Returns cannot convert. Migration: `supabase/migrations/20260828200000_pos_sale_invoice.sql` .

POS register: A register is a **till identity**, not a second sales ledger. It belongs to the organization and a **brand** (`public.companies` — the same trading identity as `invoices.company_id`). Paidly has **no locations / stores table**; do not invent a multi-site model. Fields: `name` , `status` (`active` \| `disabled`), `assigned_staff_id` (org member or owner — operational, not a second ACL), `opening_balance` (default cash float; not POS sale money and not invoice `payments`). Native checkout stamps `pos_sales_events.register_id` and **`company_id` from the register only** (request body `company_id` is ignored so a till cannot spoof another brand). The device remembers the selected till in `localStorage` . Settings → Integrations → POS registers. Yoco/Square ingress can omit `register_id` . Do **not** duplicate `pos_sales` / `pos_payments` tables. Migration: `supabase/migrations/20260828210000_pos_registers.sql` .

POS catalog (multi-brand): Still one table (`public.services`). Optional `services.company_id` → `companies.id` (`ON DELETE SET NULL`). **Null = org-shared** (visible on every till). **Set = private to that brand.** `GET /api/pos/catalog` and native checkout load `item_type = product` for the org where `company_id IS NULL OR company_id = register.company_id` . A brandless register sees shared products only. Company A’s till cannot sell Company B’s private products — the header brand switcher is not a catalog scope. Returns restock against the original sale’s lines without re-checking brand (the original sale already passed the filter). Migration: `supabase/migrations/20260828240000_pos_multibrand_catalog.sql` .

POS session (cash drawer / shift): A register session is **not** Auth `getSession` . Table `pos_register_sessions` : one **open** shift per register (partial unique index). Lifecycle: OPEN (cashier starts shift, enters opening cash) → sales and returns stamp `pos_sales_events.session_id` → CLOSE (count closing cash). Tracked: `opening_balance` , `cash_sales` , `cash_refunds` , `expected_cash` (`opening + cash sales − cash refunds`), `closing_cash` , `variance` (`closing − expected`). Card / digital tenders do not move expected drawer cash. While open, cash totals are computed live from `pos_sales_events` ; at close they are snapshotted. Closed rows are **immutable** (no PATCH; database trigger rejects UPDATE/DELETE). Native checkout/return requires an open session (`422 SESSION_REQUIRED`) unless the sessions migration is missing (degrade, do not block). External Yoco/Square ingress may omit `session_id` . APIs: `POST /api/pos/sessions` , `GET /api/pos/sessions` , `GET /api/pos/sessions/:id` , `POST /api/pos/sessions/:id/close` . Settings shows read-only closed-shift history. Migration: `supabase/migrations/20260828220000_pos_register_sessions.sql` .

POS returns / refunds (append-only): Reversing a till sale never deletes or rewrites the original `pos_sales_events` row. Flow: original sale → sale item (`items` jsonb, matched by `product_id` ; `line_id` stamped for later) → return event (`sale_kind = return` , `parent_event_id`) → inventory `in` via `adjust_inventory_stock` (`source = pos`). Original `total_amount` / `items` stay the sale; `refund_status` (`none` \| `partial` \| `full`) and `refunded_amount` are snapshots of child returns. **V1 money:** `refund_rail = till_cash` takes cash out of the open drawer (session `cash_refunds`). Card / digital restock goods with `refund_rail = pending_provider` and copy `original_payment_intent_id` ; they do **not** mark the original `payment_intents` row `refunded` and do not call Ozow or a card terminal. Staff may opt into `refund_as_cash` to pay a card/digital return from the drawer. **Future (columns exist, no API):** a new `payment_intents` row with `refund_of_intent_id` pointing at the original paid intent, then `refund_rail = provider` . Do not add `pos_sale_items` , `pos_refunds` , or dual-write invoice `payments` / SaaS `payment_history` . Permission: `pos_refund` . Migration: `supabase/migrations/20260828230000_pos_return_audit.sql` .

POS audit trail: Completed till transactions are not silently edited or erased. Money identity on `pos_sales_events` stays immutable (existing trigger). Disconnecting Yoco/Square **SET NULL**s `connection_id` — it does **not** delete sales. Lifecycle is an org-scoped append-only stream `pos_audit_events` (same pattern as `document_events` / `subscription_events` , not platform `audit_logs`): `sale_created` , `payment` , `completion` , `refund` , `cancellation` , `inventory_movement` . Unpaid checkout logs `cancellation` on the `payment_intents` row (no fake sale). Returns log `refund` on the **original** sale (`metadata.return_id`). Inventory still lives in `inventory_movements` . `GET`

`/api/pos/sales/:id/audit` (`pos_access`) returns the timeline; the receipt dialog shows it. Missing table must not block checkout. Migration: `supabase/migrations/20260828250000_pos_audit_trail.sql`.

POS offline (V1 — not a full offline till): Retail POS needs a live connection to record money. The existing **invoice SyncEngine** queue (`CREATE_INVOICE` / `UPDATE_CLIENT` in `localStorage`) is **not** a POS outbox: checkout is `POST /api/pos/checkout` (payment intent, open shift, inventory RPC). V1 does **not** enqueue cash or card sales, does **not** store payment credentials on the device, and does **not** mark card/digital paid while offline. The till **always shows** connectivity (Online / Reconnecting / Offline). Cart build and device hold (`sessionStorage`) still work offline; Pay, cash take, card, digital, returns, shift open/close, and tax-invoice copy require Online. Cash queue is a future dedicated till outbox — not the invoice sync queue — and must never look like a completed `pos_sales_events` row until the server confirms.

POS staff permissions: Till access uses the existing org membership RBAC (`src/lib/companyPermissions.js` / `server/src/companyRouteAccess.js`) — Supabase Auth + `memberships` roles `employee` / `manager` / `admin` (owner maps to admin). Do **not** add a second POS login, JWT, or role table. Grants: `pos_access` (open till, catalog, today's sales, receipts, sale audit timeline), `pos_sell` (checkout, open shift, optional tax-invoice copy), `pos_discount` (cart-level discount), `pos_refund` (returns), `pos_close_register` (close shift / cash-up), `pos_view_reports` (closed-session history and non-today sales lists). Employee: access + sell. Manager and admin: all six. Register CRUD stays `manage_company_settings`. `assigned_staff_id` on a register is operational, not a second ACL. There is still **no** `MANAGE_POS_MANUAL_CARD`. APIs return `403 POS_FORBIDDEN` when the grant is missing. `/POS` is wrapped in `RequireCompanyPermission` (`pos_access`) plus FeatureGate `pos` (Business+; first-class plan key, not aliased to inventory). SQL helper `org_has_pos_permission(org_id, permission)` mirrors those grants for RLS.

POS staff invite (till-only link): From the **front-facing till** (`/POS`), a manager or admin (`manage_employees`) can **Invite staff**. That uses the existing company invite API (`POST /api/company/invite`) — not a second till login. The invite is always RBAC `employee` plus `memberships.job_function = pos` (`source = pos`). Cashiers cannot invite others. After invite, the till shows a **special shareable link** `/invite?token=…&next=POS`. Accepting it (and `accept_company_invite_token`) copies `job_function` onto `memberships`. POS-only staff land on `/POS`, have no back-office nav (invoices, dashboard, settings), and the till back-arrow is Sign out. Enforcement is the job function, not the URL. Settings → Team can still pick job function **POS only**. Migration: `supabase/migrations/20260828290000_pos_staff_invite.sql`.

POS URL / access (back-office entry): The till is only `/POS` (alias `/pos`). Staff reach it from **Paidly navigation**, not from chrome on the till itself. Primary sidebar Overview order: **Dashboard → Invoices → Quotes → Clients → Products → POS** (then Documents, Purchase Orders, Reports, Settings). The **POS** item is shown only when the org **opted into POS** (`organizations.business_type` is `retail` or `mixed`), the plan includes the `pos` feature (Business+), **and** the user has `pos_access`. Service orgs (and unset type) never see POS in the sidebar — it is not an upgrade teaser. Company members also get **POS** as a Me-section workspace link when those same gates pass. Dashboard **Open POS** stays visible even when there are no sales today. Mobile bottom nav stays Home / Invoices / FAB / Clients / Menu; POS lives in the sidebar Menu drawer. `canShowPosNav` (`src/lib/posNavAccess.js`) is the single visibility helper.

Business type (POS is optional): Paidly is not a POS product that happens to invoice. Every tenant is a **service**, **retail**, or **mixed** business (`organizations.business_type`; NULL = service). **Service** → normal Paidly (invoices, quotes, clients) — no till. **Retail / product** → Paidly + POS. **Mixed** → invoices plus a till for walk-in sales. Invoices stay available for every type (retail is still Paidly). Onboarding and Settings → Company Profile pick the type; Settings → Integrations shows registers only when POS is on. APIs return `403 POS_NOT_ENABLED` for checkout/registers/connections when the type is service. Existing orgs that already have `pos_registers` or `pos_sales_events` are backfilled to `mixed`. Migration: `supabase/migrations/20260828270000_organizations_business_type.sql`. Helper: `shared/businessType.js`.

POS feature entitlement: Till enablement uses the existing SaaS feature map — not a second flag table and not hard-coded user or org IDs. Canonical key `pos` (`POS_PLAN_FEATURE` in `shared/planFeatures.js`) sits on **Business+** (`FAMILY_FEATURES`, same tier as inventory but a separate key so POS can be packaged independently later). Server till operations call `requirePosPlan` → `requireFeature('pos')` after membership/RBAC and before business-type (`requirePosCapability`). That covers catalog, checkout, return, registers, sessions, connections, POS `payment-intents`, receipt email, and tax-invoice copy. **Do not** gate GET sales lists, Cash Flow, or provider webhooks on `pos`

— those are historical money and callbacks. SPA uses `hasFeature` / `hasFeatureAccess('pos')` / FeatureGate; nav hides POS when the plan lacks `pos` rather than showing an upgrade teaser. Catalog display: append `pos` to `plans.features` jsonb (`supabase/migrations/20260828280000_plans_pos_feature.sql`). Enforcement SoR remains subscriptions family → `FAMILY_FEATURES`, not the jsonb list and not `profiles`.

POS acceptance tests (Task 28): The till contract is `tests/unit/posTask28Acceptance.test.js` (TESTs 1–15: open till, search, cart, quantity, cash change, digital paid vs not paid, inventory commit point, receipt, attach customer, register brand, RBAC, persistence vs held cart, tenant isolation, duplicate payment). Supporting modules: `tests/unit/posSaleProcessor.test.js` (webhook idempotency), `tests/pos.spec.ts` (guest cannot open `/POS`; entitled session may exercise the till). Do not treat a held `sessionStorage` cart as a sale. Digital checkout writes `pos_sales_events` only when `posSaleCompletesWhenPaid` (`payment_intents.status = paid`).

POS tenancy, RLS, and money identity: Every POS row is `org_id`-scoped. Org members may **SELECT** sales, sessions, registers, connections, and audit events (Cash Flow / Reports read `pos_sales_events` under RLS). **Writes** of sales, sessions, registers, connections, audit events, and `payment_intents` are **service_role / API only** — the browser must not insert a sale. Triggers reject cross-tenant FKs: register `company_id` and sale `company_id` must be a `companies` row in that org; sale `client_id`, `register_id`, `session_id`, `connection_id`, and `parent_event_id` must match `org_id`. Native checkout does **not** trust client totals: catalog price × quantity, server discount (capped, `pos_discount`), tax_amount 0 on the till, payable total written to `payment_intents.amount` and `pos_sales_events.total_amount`. Client `unit_price` / `total` / `tax_amount` / `company_id` are ignored or rejected on mismatch. Cash tendered is validated against that server total; change is computed on the server. Yoco/Square webhook adapters still ingest the provider's completed amount (external till). Migration: `supabase/migrations/20260828260000_pos_rls_tenant_integrity.sql`. To apply the full native POS chain in the SQL Editor in one paste, use `scripts/apply-native-pos.sql` (idempotent; starts with webhook POS tables if they are missing).

Schema (`payment_intents`):

- `id`, `org_id`, `source_kind`, `document_id` (nullable), `pos_sale_event_id` (nullable), `provider`, `amount`, `currency`, `status` (includes unused-in-V1 `refunded`), `external_id`, `idempotency_key`, `refund_of_intent_id` (nullable; future provider refund pointing at the original paid intent), `created_at`
- Migrations: `supabase/migrations/20260828180000_payment_intents.sql`, `supabase/migrations/20260828190000_payment_intents_card_terminal.sql`, `supabase/migrations/20260828230000_pos_return_audit.sql`, `supabase/migrations/20260828250000_pos_audit_trail.sql`, `supabase/migrations/20260828260000_pos_rls_tenant_integrity.sql`

Boundary of ownership:

- **Document Engine owns:** payable snapshot creation, due/expiry semantics, and `document_ref` identity.
- **Native POS owns:** cart snapshot (no stock movement), **till cash** (tendered + change, independent of online rails), writing `pos_sales_events` only after the intent is `paid`, then inventory via `adjust_inventory_stock`.
- **Revenue & Ops owns:** provider orchestration, webhook verification, settlement/reconciliation. Invoice settlement still uses `payments`; SaaS billing still uses PayFast ITN → `payment_history`.
- **Experience System owns:** one payment-status language and CTA behavior across document detail, public links, portal views, and the till Pay sheet.

Result: no direct "mark paid" shortcuts from UI paths; all payable settlement flows through `Payment Intent` + verified payment events. Ozow is the first intended customer online rail; the provider interface is registered now — full Ozow charge/webhook wiring waits on merchant credentials.

Product upgrade: one compose surface, many kinds

The **big upgrade** is not more list pages—it is **one mental journey** with honest persistence:

1. **Create** from a consistent compose pattern (brand, line items, preview).
2. **Commercial types** (invoice / quote / payslip) write to specialised tables; **other types** write to the Documents Hub.
3. **Shared UI** where it already exists — editors, preview, send affordances (Experience System + templates).

4. **Different logic** — status machines, tax/settlement rules, payroll vs AR: **kind-specific adapters** behind the same surface.

List routes ("Invoices", "Quotes", "Payslips") remain **indexes** over their specialised tables. The Documents Hub lists generic `documents` rows only.

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5. Relationship System

Job: Who you sell to and **what** you sell—data that **feeds** the Document Engine.

- **Clients** (CRM): contact, terms, portal access
- **Catalog** (`services`): products & services, pricing, inventory where relevant
- **Line items** on documents: snapshots + links back to catalog when useful
- **Native POS till** (`/POS` or `/pos`): a **front-facing till** (not the admin dashboard). Layout skips sidebar/header/footer; the page is catalog + cart + pay. Staff open it from back-office **POS** nav (after Products) when inventory and `pos_access` allow — the till has no app chrome of its own. Search is scan-first: USB/Bluetooth wedge or camera barcode → catalog lookup (name, SKU / product code, barcode) → cart. Customer is optional (**Walk-in Customer** default); staff can search/select or create on the same `clients` table without leaving the till. Reads the same catalog and customers; it does **not** own a second product or client store. Back office remains the master for products, stock, pricing, and customers.

Strategy: Treat this as the **input graph** to commerce—not a separate "Contacts app." List screens are views; the system is **relationship + SKU/rate intelligence**. Growth-oriented **CRM behaviour** (follow-ups, nudges, sequences) lives primarily in the **Growth Engine**; this system owns **master data** and what gets embedded on documents.

One catalog (hard rule): `public.services` is the only product/service master (`item_type`: `product` | `service` | `labor` | `material` | `expense`). Purchase orders, deliveries, `inventory_movements`, invoices, and POS all resolve to `services.id`. Optional `services.company_id` scopes **private** brand products for POS; it is not a second catalog. Do **not** revive `public.products`, add `products_v2`, or a parallel SKU store. Legacy `public.products` / `stock_transactions` were dropped in `supabase/migrations/20260804121000_inventory_movements_fk_and_legacy_cleanup.sql`. `entities/Product.json` is unused schema — do not wire a Product entity against a second table.

Known gaps (real, but not a second catalog):

- **No product variants** — one SKU and one barcode per `services` row. A child table is justified only if retail needs size/color SKUs, and it must still parent to `services`.
- **No `pos_sale_items` / `pos_refunds` tables** — sale lines stay jsonb on `pos_sales_events`. Checkout merges by `product_id`; returns match those lines. `line_id` is stamped for a later split-line model. Till lifecycle is `pos_audit_events`, not a second sales ledger and not platform `audit_logs` / `document_events`.
- **POS provider refunds are not V1** — Ozow / `card_terminal` money-back is not called. Return events stamp `refund_rail = pending_provider` and `original_payment_intent_id`. Future: `payment_intents.refund_of_intent_id` + `refund_rail = provider`. Cash returns (`till_cash`) are V1.
- **No category master** — `services.category` is free text. A later `categories` table would FK from `services`, not from a new products table.
- **No locations / stores table** — a POS **register** (`pos_registers`) is a till identity on the org + brand (`companies`). Catalog visibility is `(services.company_id IS NULL OR services.company_id = pos_registers.company_id)`. It is not a sales ledger and not a multi-site model. Sale SoR stays `pos_sales_events` (`register_id`).
- **No offline POS ledger** — V1 till checkout is online-only. Invoice SyncEngine is not used for POS. Cart hold is device `sessionStorage` only. Do not fake completed sales while offline.
- **No online store / cart / e-commerce orders table** — till cart is device `sessionStorage`; B2B sale is `invoices`; retail sale is `pos_sales_events`. Do not add a generic `orders` table that dual-writes invoices.
- **`payment_intents` exists** as the customer handoff (POS cash / Ozow / `card_terminal`, document Ozow). **Settlement stays three rails:** invoice `payments`, SaaS `payment_history`, POS `pos_sales_events`. Do not merge those into one payments table. PayFast never appears on `payment_intents`.

- **Reports / Accounting / Cash Flow** read invoices, expenses, `payments`, and `pos_sales_events` (`src/utils/cashFlowTruth.js`, `src/utils/posSalesTruth.js`). Optional POS tax-invoice copies (`invoices.pos_sale_event_id`) are excluded from invoice income so converting a sale does not double-count till cash.

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4. Revenue System (Revenue & Ops)

Job: Everything that turns **issued documents** (especially invoices) into **cash, visibility, and tenant billing**—without re-implementing document authoring here.

- **Payments:** Payfast, invoice payment state, webhooks (`api/payfast-handler`)
- **POS (native till + integrations):** Paidly POS (`/POS`) is a front-facing checkout on the **same** catalog, inventory, customers, and `pos_sales_events` table — not a separate app or database. Checkout creates a `payment_intents` row. Cash is settled on the till (tendered + change). Digital Payment uses `ozow` and is completed only after provider confirmation. Card is `card_terminal` — architecture-ready, not a fake terminal; the sale is not marked paid on click. `pos_sales_events` is written only after the intent is `paid`. Stock does not decrease for cart, Pay, or unpaid intents; after the sale row exists, checkout calls `adjust_inventory_stock` (`source = pos`). Returns go through `POST /api/pos/return` (session auth, `pos_refund`, `pos` plan entitlement): append-only `sale_kind = return` against `parent_event_id` — the original sale is not deleted. Stock moves only via `adjust_inventory_stock` (`source = pos`). Cash returns use `refund_rail = till_cash`. Card/digital V1 restocks with `pending_provider` (no Ozow/terminal refund). Checkout never creates an invoice. Optional **Customer requests invoice** (`POST /api/pos/invoice`) writes a paid tax-invoice copy linked by `invoices.pos_sale_event_id` — not a new receivable and not invoice `payments`. PayFast is not used for POS. External tills (Yoco/Square/generic webhook) remain **adapters** into the same sale stream. Append-only `pos_audit_events` records sale created, payment, completion, refund, cancellation, and inventory movement; disconnecting a POS connection does not delete completed sales. Dashboard **POS sales today**; Settings → Integrations for registers and hardware connect. Migrations: `supabase/migrations/20260828160000_native_pos_checkout.sql`, `supabase/migrations/20260828180000_payment_intents.sql`, `supabase/migrations/20260828190000_payment_intents_card_terminal.sql`, `supabase/migrations/20260828200000_pos_sale_invoice.sql`, `supabase/migrations/20260828210000_pos_registers.sql`, `supabase/migrations/20260828220000_pos_register_sessions.sql`, `supabase/migrations/20260828230000_pos_return_audit.sql`, `supabase/migrations/20260828240000_pos_multibrand_catalog.sql`, `supabase/migrations/20260828250000_pos_audit_trail.sql`, `supabase/migrations/20260828260000_pos_rls_tenant_integrity.sql`, `supabase/migrations/20260828270000_organizations_business_type.sql`, `supabase/migrations/20260828280000_plans_pos_feature.sql`.
- **Payment intents:** capture/retry/expiry orchestration and provider status normalization
- **Cash flow:** timelines and balances built from **settled invoice payments** (income), **completed POS `pos_sales_events`** (till sales in, till-cash refunds out), and recorded expenses, plus open invoice remainders for outstanding/upcoming cash. The Cash Flow page paginates entity lists so totals are not capped at the default 100-row fetch. Paid invoices with `pos_sale_event_id` are skipped so an optional tax-invoice copy does not appear as invoice income; the till event is the cash-in row. Card/digital `pending_provider` returns restock goods but do not reduce cash-flow income until a provider refund exists.
- **Reports:** same cash-flow truth helpers as Cash Flow (`src/utils/cashFlowTruth.js` + `src/utils/posSalesTruth.js`) — settled invoice payments and till sales in, recorded (non-rejected) expenses out, calendar-day periods. Paginated fetches; do not dual-write Documents Hub expense *claims* into `expenses`. POS retail metrics (today, cash/digital, refunds, top products, units, gross/discount/tax/net) are a **view** over `pos_sales_events`, not a second reporting engine.
- **Subscriptions & dunning:** how Paidly bills the customer (packaging, crons in `vercel.json` → `/api/cron/...`)
- **SaaS subscription SoR (billing v2):** `plans` catalog (`amount`, `billing_cycle`, `plan_family`, `payfast_item_name`, `features`, `active` — **not hardcoded in React**) + public tiers Starter R50 / Business R150 / Growth R350 (+ annual 2 months free) + Enterprise contact-sales; legacy Individual/SME/Corporate grandfathered.

`subscriptions` core (`company_id`, `plan_id`, `plan_family`, `grace_ends_at`, PayFast ids, period bounds, `cancelled_at`, `created_by`, `trial_started_at` / `trial_ends_at`, `subscription_source` (`system_trial` \| `payfast` \| `admin`), `admin_override`; status allow-list only: `pending` \| `processing` \| `active` \| `past_due` \| `failed` \| `cancelled` \| `expired` \| `suspended` \| `trialing`) + append-only `payment_history` + `subscription_events` + `payfast_itn_logs` + `subscription_invoices` + `webhook_logs`. **New org-owner accounts created on/after 2026-08-20 00:00:00 UTC** get a **7-day server-side trial** (`trialing`); invited members do not. Access is `hasSubscriptionAccess` (trial not expired, `active`, or admin-managed). Activation to paid only after verified PayFast ITN (never from the SPA). Admin PATCH `/api/admin/subscriptions` can extend/grant/suspend/cancel and always sets `admin_override` so expiry jobs cannot revert it; changes write `audit_logs`. **Payment/revenue reporting** counts only `payment_history.payment_status = completed` with transaction/created time **≥ 2026-08-20T00:00:00.000Z** (UTC) — not trials, admin grants, pending/failed, or pre-cutoff rows. **APIs:** `GET /api/subscriptions/plans`, `POST /api/subscriptions/create|change|cancel`, `GET …/status|current`, `POST /api/payfast/itn`, `GET /api/admin/subscriptions` (overview + reporting), `GET /api/admin/payments`, `PATCH /api/admin/subscriptions` (legacy client-priced `POST /api/payfast/subscription` returns 410). Schema: `docs/SUBSCRIPTION_BILLING_SCHEMA.md`. Profiles (`plan` / `subscription_status` / `is_pro`) remain a cache (mirrored as `plan_family`).

Feeds off the Document Engine: revenue truth is **downstream of document state** (totals, status, due dates, line items). Revenue & Ops aggregates, reconciles, and projects; it does not fork “another invoice.”

Strategy: Keep business rules that define **what a document is** (e.g. when an invoice is *overdue* in product terms) aligned with the Document Engine; keep **money rails** (capture, allocate, subscription charge) here.

Payment integration rule: provider events never mutate document totals directly. They update payment-intent/payment records first; document settlement status is derived via verified reconciliation rules. SaaS tenant billing is separate from invoice `payments`: ITN must verify signature, PayFast server confirm, amount, merchant, and subscription correlation before moving `subscriptions` off `pending`.

SaaS billing security principles (non-negotiable): (1) Never trust the frontend for payment state. (2) Verify every ITN server-side with PayFast before changing subscription status. (3) Database is SoR. (4) Log every lifecycle event and webhook. (5) Idempotent webhooks (no duplicate activations). (6) RLS everywhere; service role only where required. (7) Never expose merchant credentials, signatures, or verification logic to the client. (8) Validate merchant ID, amount, currency, and subscription identifiers before activate. (9) Prefer atomic DB transactions for payment history + subscription update + event logs. Details: `docs/SUBSCRIPTION_BILLING_SCHEMA.md`.

Responsibility split (hard boundary)

Document Engine owns:

- Document status semantics (`paid`, `overdue`, lifecycle transitions)
- Document totals and payable snapshots

Revenue & Ops owns:

- Payment providers (Payfast, Stripe, future rails)
- Subscriptions and dunning/billing operations
- Cash flow models
- Reporting/read models

Boundary outcome: clear separation keeps document rules coherent while allowing payment/billing systems to scale independently.

7. Payment Intelligence Layer (Get Paid System)

Job: Turn events + payment intent state into proactive actions that improve collection velocity.

- Inputs: `document_events`, `payment_intents`, due dates, client/payment history
- Decisions: reminder timing, CTA sequencing, retry windows, provider fallback policy
- Outputs: auto reminders, smart nudges, “invoice viewed but not paid” triggers, suggested follow-ups, prioritized “at-risk” invoices, and payment funnel analytics

Example trigger (v1):

Client viewed invoice 3 times without a `paid` event in the observation window → trigger reminder and surface a suggested follow-up action in the right panel.

Position in architecture: sits between observation/payment state and user-facing actions, orchestrating how Paidly gets customers paid faster without leaking provider logic into page code.

Growth loops (implemented within Revenue, Relationship, and Experience systems)

Job: How Paidly **acquires, retains, and scales**—loops that sit beside day-to-day issuing.

- **Emails:** transactional and campaign-style sends from `/api` and app-triggered flows
- **Notifications:** in-app + scheduled nudges (crons, reminders, due-date services)
- **CRM behaviour:** follow-ups, client engagement, portal nudges—**behaviour** on top of Relationship data, not a duplicate CRM product

This is how Paidly scales: repeatable growth mechanics without entangling them in the document compose path.

Operator / platform admin (users, oversight, platform messages, `/admin-v2/*`) can be documented as part of this engine or Identity, depending on audience—treat it as **platform operations**, not SMB document logic.

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6. EXPERIENCE SYSTEM (UX AT SCALE)

Job: So Paidly **feels** like one product, not twenty features stitched together.

Pillars:

Pillar	Meaning
Shell	Repeated editor layout: full-width work area, `max-w-*` content, sticky summary/actions (pattern already emerging: EditQuote, EditClient, EditCatalogItem).
Tokens	`border-border`, `bg-card`, `text-muted-foreground`—no one-off palette per page.
Data discipline	React Query keys, invalidation rules, **cache-first navigation** (`paidlyClientCachePolicy.js` + `docs/Paidly-Caching-Architecture.md`). Hydrate from Zustand/Dexie/RQ then refresh when stale — not a full cold load on every route entry. **Observability:** `paidlyDataLayerInstrumentation.js` (cache restore, dedupe, realtime patches) + `[PaidlyRealtime]` structured logs (reconnect/backoff/cooldown).
A11y & forms	Label/`id` parity, focus order, disabled states that explain *why* (title/tooltip).
Money UX contract	Same status chips + primary CTA logic (`Pay now`, `Retry payment`, `View receipt`) everywhere a payable document appears.
Page template	List/index pages share one **three-zone** shell (below)—visual consistency reads as **premium**.
Installable app	Same SPA, installable as a PWA (`display: standalone`). Service worker caches the **application shell** (HTML/JS/CSS/icons) only. **Never** cache invoices, quotes, payslips, clients, payments, or Supabase/`/api` responses. Auth stays on live GoTrue + `localStorage` (`paidly-auth`).

Strategy: Treat "Experience" as **governed**: a short **layout + form checklist** for any new surface that touches Identity or the Document Engine—same as you'd gate API changes.

Installable PWA (same codebase, same auth): Chromium can show **Install Paidly** (`beforeinstallprompt`). iOS/iPadOS uses **Share → Add to Home Screen**. Launch uses `display: standalone` (no browser chrome). The service worker precaches the shell and uses **NetworkOnly** for Supabase and `/api`. A waiting worker surfaces an Update toast — it must not force-reload while the user is editing a document. This is not a second desktop app (no Electron/Tauri).

Experience upgrade: make the editor pattern universal

The editor shell is a product advantage only if it is consistent across **documents and money actions**. Standardize the same right-side sticky panel pattern for invoice payment actions, not just compose/edit forms.

Universal right panel contract (sticky):

- Primary amount block (`Total`, optional secondary currency conversion).
- Primary action first (`Pay Now` when payable).
- Secondary lifecycle actions below (`Send Reminder`, receipt/share actions by state).
- Deterministic state switching from `Payment Intent` status (no page-specific CTA ordering).

Invoice payment panel (reference pattern):

- `Total: $100`
- `≈ R1,638`
- `[ Pay Now ]`
- `[ Send Reminder ]`

Page Template System (UI system — critical)

The UI is improving but must be **systemized**, not page-by-page improvisation.

Every primary list / index page follows three zones:

1. **Header** — page title, one-line purpose, **primary actions** (create, import, refresh, layout toggle). Prefer `PageHeader` (`src/components/dashboard/PageHeader.jsx`) inside `PageTemplate.Header`.
2. **Content** — main **table or grid** (or tabbed main). This is the scroll-heavy region; keep widths `min-w-0` so tables don't blow the shell.
3. **Side panel** — **filters**, **counts / summary**, secondary controls. On large screens: **sticky** column (`lg:`) to the right of content; on small screens: stack **below** main or **above** the table (pick one per product area and stay consistent).

Code: `PageTemplate` + `PageTemplate.Header` + `PageTemplate.Body` (`sidePanel` prop) in **`src/components/layout/PageTemplate.jsx`**. Import from `@/components/layout` or `@/components/layout/PageTemplate`. Use **`embedded`** when the page already sits inside a padded shell (e.g. **`AdminLayout`**).

Conformance targets (same chrome, same rhythm):

Page	Scope
Invoices	Primary document list — header/actions + table; filters/summary → side panel
Quotes	Same pattern as invoices
Clients	Relationship index — header + list/grid; filters or segment summary → side
Services	Catalog / inventory — header + content; industry/templates/search → side where possible

Refactors can be **incremental**: introduce `PageTemplate` first, then move filters/summary into `sidePanel` per page without changing data logic.

Standardize UI layout (everything same structure)

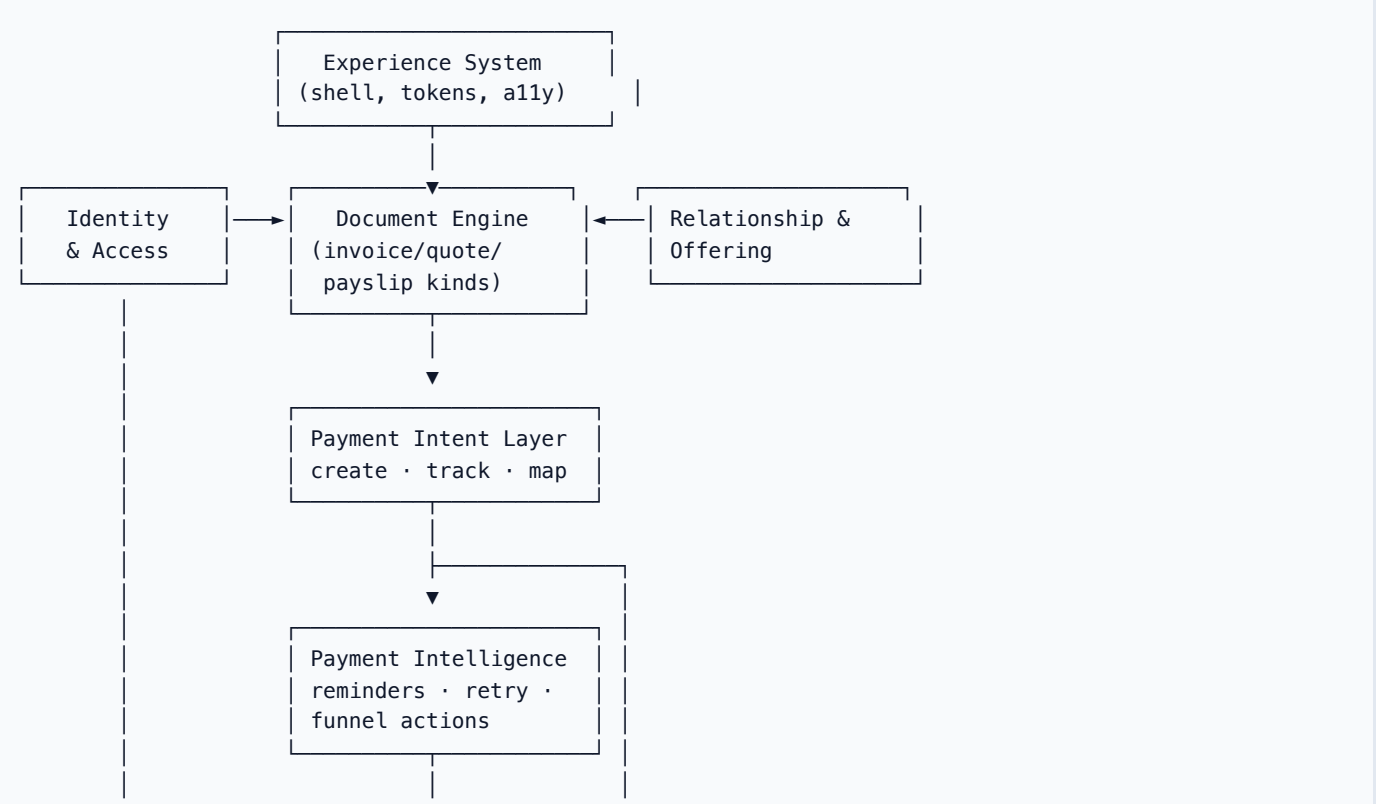
Rule: major surfaces share one **structural grammar** so the product reads as intentional, not a stack of one-off pages.

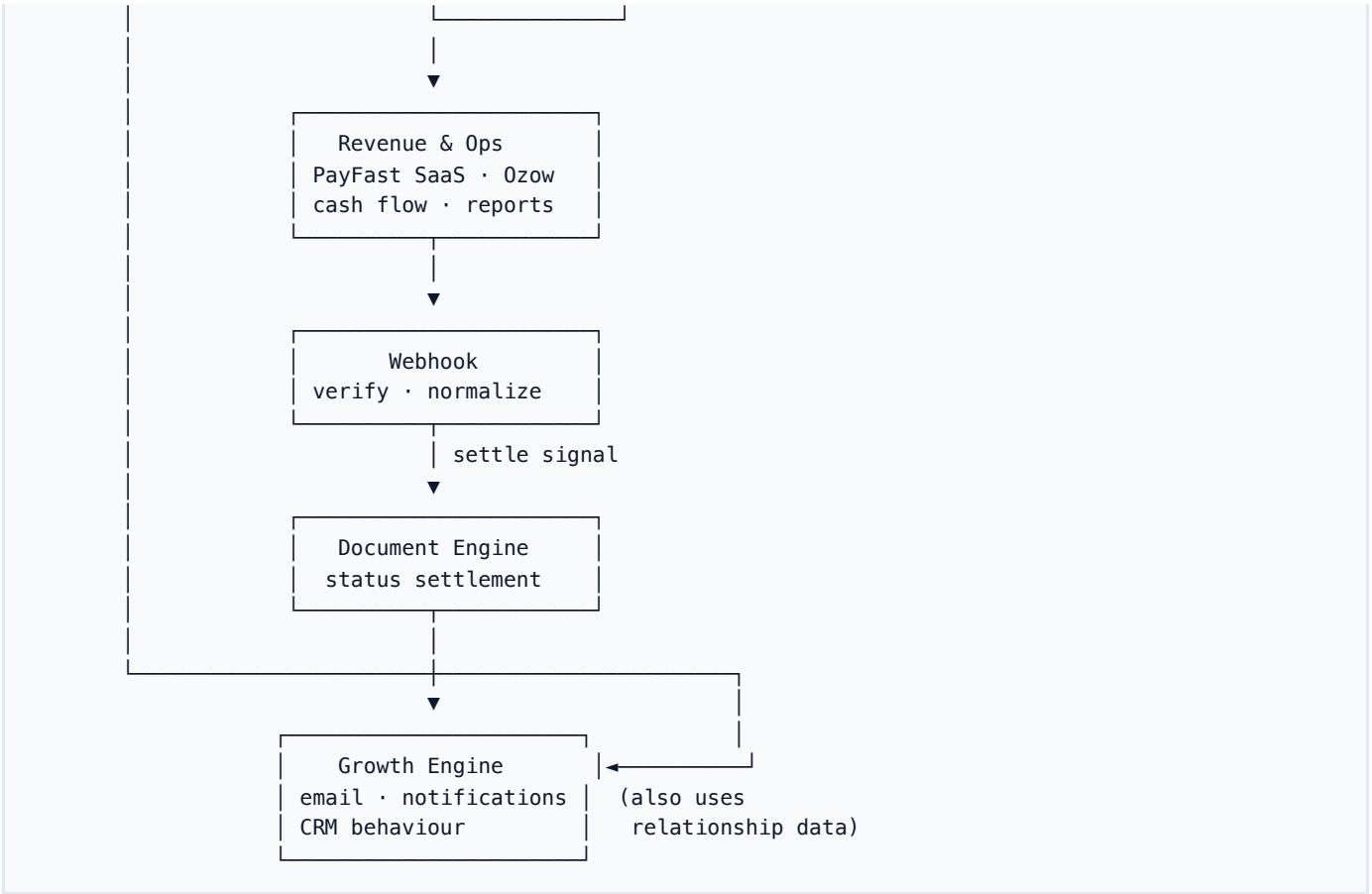
Surface type	Structure
List / index (Invoices, Quotes, Clients, Services, …)	**`PageTemplate`**: Header (title + actions) → Content (table/grid) → Side panel (filters / summary)

Surface type	Structure
Document compose / edit	Shared **editor shell**: full-width work area, `max-w-*` content, **sticky** summary + primary actions (align `EditInvoice` / `EditQuote` / `CreateDocument` patterns)
Payment touchpoints (document detail, public invoice, portal)	Shared payment state model driven by `Payment Intent` status; same CTA order and error/retry messaging
Retail POS till (`/POS` , `/pos`)	**Dedicated till shell** — not `PageTemplate` . No dashboard sidebar, header, footer, or mobile bottom nav. Staff enter from back-office **POS** (sidebar after Products; company workspace Me link; Dashboard Open POS). Full-viewport catalog + cart, large touch targets, scan-first search. Header always shows **connectivity** (Online / Reconnecting / Offline) and selects a **register** (till identity on a brand; opening float shown; catalog is that brand's shared + private products only). Cart: add/remove, qty ± / edit, cart-level discount (invoice identity: listed price due; tax not invented on the till), clear, device hold in `sessionStorage` (not a sale). **Checkout requires Online** — cash is not queued; card/digital never complete offline. Cart and Pay do not decrement stock. Inventory moves only after verified payment writes `pos_sales_events` , via `adjust_inventory_stock` . Customer is optional: default **Walk-in Customer**; search/select or create on the existing `clients` table without leaving the till. Pay: **Cash** (till-verified), **Card** (architecture-ready, not click-to-paid), **Digital Payment** (Ozow; paid only when Ozow confirms). After a completed sale: print / download / email a **receipt** (not an invoice) branded from the **register's** company, not the header document-brand switcher. The receipt dialog shows the append-only audit timeline. Optional **Customer requests invoice** creates a paid tax-invoice copy of that sale (named customer required) — it does not ask them to pay again. **Return** (managers, `pos_refund`): pick remaining qty per original line; original sale stays; cash leaves the drawer; card/digital restock with pending provider refund. Back office remains the product/customer master.
Admin / settings-style	**`PageHeader`** + main column; use **`PageTemplate`** **with** **`embedded`** when inside `AdminLayout` if a side rail helps

Deliverable: keep the **Experience checklist** short: "Which template? Header actions? Side panel? Sticky save?"— **every new page picks a row**, no ad hoc fourth layout.

A.3 SYSTEM MAP (ONE PICTURE)





Money loop (explicit):

Customer money and SaaS billing are **completely separate domains**.

`Document Engine` / `POS till` → `Payment Intent Layer` → customer rail (`cash` | `ozow` | `card_terminal`) → verified event → settle (`payments` for invoices, `pos_sales_events` for till).

SaaS billing (not a customer rail): Paidly subscription → PayFast → `subscriptions` + `payment_history`. Never POS. Never till `payment_intents`.

Get Paid loop (differentiator):

`Observe` (`document_events`) + `Payment Intent` state → `Payment Intelligence Layer` → smart reminders/CTA/retry strategy → `Revenue & Ops` + `Experience System`

Retail POS loop:

`Relationship (catalog + customers)` → `POS till (select register → cart → Cash | Card | Digital Payment)` → confirmed payment intent → `pos_sales_events` (`register_id`) → `adjust_inventory_stock` (`source = pos`) → **retail receipt** (print / PDF download / email) + append-only `pos_audit_events`. Till **must be Online** for checkout (V1 does not queue cash on the invoice sync queue). Optional **Customer requests invoice** → paid `invoices` row with `pos_sale_event_id` (tax copy only; no invoice `payments`, no second stock movement). **Return:** original sale stays → append `sale_kind = return` (`parent_event_id`) → restock `in` → cash `till_cash` or card/digital `pending_provider`. Cart, Pay sheet, and unpaid intents do not move stock. Checkout does **not** write `invoices`. Cash is till settlement. Digital is Ozow (paid only when Ozow confirms). Card is not click-to-paid; Yoco/Square hardware confirms via webhook adapters. PayFast stays on SaaS billing.

Part B — Technical blueprint (stack & flow)

B.1 WHAT PAIDLY IS (PRODUCT ONE-LINER)

Paidly is a **business operating system for SMBs**—not a narrow invoicing tool: **documents** (invoice / quote / payslip), **relationships & catalog**, and **revenue visibility & payments** in one product story. **South Africa–first** (Payfast, ZAR defaults). **Stack: Vite + React** SPA on **Vercel**, **Supabase** as system of record.

B.2 WHAT RUNS WHERE

Layer	Technology	Role
Frontend	React 18, Vite 6, React Router 7	UI; lazy routes
Styling	Tailwind, Radix, Framer Motion	Components + motion
State	TanStack Query, Zustand, Context	Cache, auth shell, prefs
Data	Supabase JS	Postgres, Auth, RLS
APIs	Vercel `/api/*`	Email, shares, Payfast, crons
Optional	`server/` Express	PDF/email dev tooling
Analytics	`@vercel/analytics`	Prod only

B.3 EXTERNAL SERVICES

Service	Role
Supabase	DB + Auth (`VITE_SUPABASE_*`)
Vercel	Host + serverless + crons + redirects
Payfast	Payments / subscriptions
Resend / SMTP	Transactional email from `/api`
Anvil	PDF generation paths (tooling + app)
IP rate limits	Sign-in / sign-up / forgot-password via Node auth API when enabled

B.4 REPOSITORY MAP

- `src/pages/` — route entrypoints (thin controllers)
- `src/components/layout/PageTemplate.jsx` — **Page Template System** (header + body grid + optional sticky side panel); `embedded` for `AdminLayout`
- `src/components/` , `src/services/` — feature + domain logic; **list/query orchestration** lives in `src/services/*` (called from hooks), not in pages
- `src/api/customClient.js` — `EntityManager` → Supabase (+ localStorage for unmigrated entities)
- `api/` — Vercel handlers
- `supabase/migrations/` — schema

See **`docs/SUPABASE_DATA_MODEL.md`** for table ↔ entity detail.

For runtime ownership and boundary rules (Browser/Supabase/Edge-Server), see **`docs/ARCHITECTURE_RUNTIME_MAP.md`** .

For the connection lifecycle coordinator, semantic events, and authority rules, see **`docs/CONNECTION_LIFECYCLE_ARCHITECTURE.md`** .

B.5 DATA FLOW (REFINED)

Canonical client stack

Do **not** wire pages or heavy components **straight to** `Invoice` , `Quote` , etc. for anything beyond trivial one-offs. Prefer:

```
UI (pages / components)
  → Hooks (TanStack Query, local UI state)
    → Entity / domain service layer (`src/services/*`, orchestration)
      → Entity facades (`@/api/entities` — thin `EntityManager` API)
        → Supabase (+ RLS) / optional localStorage mirrors
```

Why the middle layer: one place for **timeouts/retries**, **logging/metrics**, **feature gating**, **error shaping**, and **swapping storage** without rewriting every screen. Hooks stay thin (cache keys, `enabled`, composition); services own **how** data is fetched or mutated.

Anti-pattern: `UI → Invoice.list()` scattered across pages.

Reference example: `useInvoices` → `fetchInvoiceListPage` in **`src/services/InvoiceListService.js`** → `Invoice.list` → `EntityManager` → Supabase. PDF/email/share flows stay in **`src/api/InvoiceService.js`** (different concern: delivery, not list reads).

Canonical profile path (implemented)

Profile state has a single fetch owner:

- **Owner:** `AuthContext` (`src/contexts/AuthContext.impl.jsx`)
- **Consumer path:** `useAuth()` and `useUserProfileQuery()` (selector facade over auth state)
- **Rule:** pages/components should not add new `User.me()` fetches for normal shell state; use auth/context selectors.
- **Allowed exceptions:** explicit one-off snapshot reads in render/send pipelines (PDF generation, email send branding snapshots) where point-in-time profile capture is intentional.

Domain-approved path matrix

Domain	Approved path
Auth/profile	`AuthContext` → `useAuth` / `useUserProfileQuery`
Document lists	hook (`useInvoices`/etc.) → service (`*ListService`) → entity facade
Document exports	page action → `DocumentExportService`
Dashboard bounded data	dashboard hooks → `DashboardDataService`
Side datasets	feature hook with explicit `enabled` gate + stale time

`customClient.js` / EntityManager — online vs offline

`EntityManager` is powerful but easy to misuse: **silent failures**, **localStorage** paths for guests, and **`list()` timeouts** returning an empty cache can look like "no data" when the network failed.

Policy (implemented in `EntityManager`): use `navigator.onLine` as a coarse gate:

- **Offline** — do **not** call Supabase for bulk `pullFromSupabase` / empty `find()`; use **in-memory / local** only and **log** (`[Paidly][EntityManager] … offline`). `get()` cache misses throw a **clear** "not available offline" error instead of a failed network round-trip.
- **Online** — **attempt Supabase** as today. When `list()` uses a **`maxWaitMs` race**, the continuation `pull` **promise** logs failures instead of `void pull.catch(() => {})`, and an **empty cache** after the wait emits a **diagnostic warning** (slow vs failed vs still loading).

`skipLocalPersistence` (signed-in + mapped table) already avoids treating **localStorage** as authoritative; the online/offline split further reduces **masked** behaviour.

Session and side effects

Auth session → org scope → the stack above + Query cache → ** `/api/` * for secrets and side effects (email, Payfast, public tokens, admin queues).

B.6 DEPLOYMENT

`vite build` → Vercel static + `vercel.json` rewrites + scheduled crons; production domains alias to the project.

HIGH IMPACT NEXT (PRODUCT — WHAT TO SHIP FIRST)

These three compound **retention**, **differentiation**, and the **business OS** story. Run them as explicit initiatives; the numbered backlog below is hygiene and scale in parallel.

1. Keep the document system honest (split persistence)

- **Invoice + quote + payslip = same engine, different tables** — shared **lifecycle** primitives (draft → send → observe → settle / convert) in `src/document-engine/`, specialised persistence (`invoices`, `quotes`, `payslips`). Do **not** migrate these into `documents`.
- **Documents Hub** — generic business documents only (`documents` / `document_items` / `document_events`).
- **Shared UI** — one **Create / Edit document** shell where it already exists for invoice/quote compose; hub types use typed/dedicated pages. Kind-specific rules stay behind thin adapters.

2. Client Timeline (inside client profile)

A single **chronological timeline** on the client record, for example:

- **Invoice sent** (and viewed / paid where data exists)
- **Quote sent / accepted / declined / expired**
- **Payment received** (linked payment rows)

Data: join `invoices`, `quotes`, `payments`, and optionally `document_sends` / `message_logs` (and later tasks/notes) filtered by `client_id` + `org_id`, sorted by time.

Why it hits: turns “contacts” into **relationship history**—high **retention** and a clear Paidly-only view most invoicing-only tools do not unify on one screen.

3. Conversion flow: Quote → accepted → auto invoice draft

When a quote moves to **accepted** (or explicit user action “Convert to invoice”):

1. **Create** an **invoice draft** (link `quote_id` or metadata for traceability).
2. **Prefill** client, line items, currency, terms/branding from the quote.
3. **Route** the user to **Edit invoice** to review, adjust tax/dates, then send.

This closes the **commercial loop** in-product. Persistence is `quotes` → `invoices`, not the Documents Hub.

Hub job cards, project reports, and scopes **Convert to Invoice / Quote** the same way: specialised compose, specialised tables.

Blueprint PDF: regenerate from this markdown with `npm run docs:blueprint-pdf` (`scripts/generate-blueprint-pdf.mjs`).

FOUNDATION PRIORITIES (RUN WITH HIGH IMPACT)

Ship these in parallel with **High impact next**—they reduce churn and make every subsequent feature cheaper.

4. Stabilize Auth + Session (work already started)

- **Goal:** predictable **sign-in, refresh, tab sync, and logout**; no ghost states where the UI runs but org or profile is wrong.
- **Scope:** Supabase Auth session lifecycle, **`getSession` / `getUser`** ordering and retries (see patterns in **`customClient.js`**), **`RequireAuth`** and role gates, **org bootstrap** (`ensureUserHasOrganization`), **invite / reset-password** flows, **profile** load after cold start.
- **Exit criteria:** short **runbook** (or ADR) for “session failure modes + what the user sees”; fewer uncaught `AbortError` paths; QA checklist for multi-tab and slow network.

5. Standardize UI layout (same structure everywhere)

- **Goal:** one **layout grammar** across the app (see **A.2 — Standardize UI layout** table).
- **Scope:** roll `PageTemplate` through primary lists; align **document editors** on the same shell; admin/settings use `embedded` + `PageHeader` where applicable.
- **Exit criteria:** published **Experience checklist** (one page) that references `PageTemplate` and editor shell; new PRs default to an existing template row—no orphan layouts.

NEXT STEPS (STRATEGY → ENGINEERING BACKLOG)

*Numbering map: **4–5** execute **Foundation §4–5** (auth/session, UI layout)—the same priorities called out in architecture **A.1** and **A.2**. **6** (Experience checklist) **closes** Foundation §5. **7** (role matrix) **supports** Foundation §4. Items **8–11** are the former 6–9 line-up (**PageTemplate** rollout, hook→service pattern, Client Timeline, quote→invoice). Product-led **High impact** items (§1–3) stay in that section above.*

1. **Grow** `src/document-engine/` — status enums, send + PDF adapters, and thin facades over `Invoice` / `Quote` / `Payslip` where behaviour overlaps (**supports § High impact 1**).
2. **Line up quote / invoice / payslip** on the same **deliver + observe** interfaces (even if tables stay separate short term); keep **compose** converging on **Create Document + type**, not parallel product UIs (**supports § High impact 1**).
3. **Add first-class Payment Intent model + APIs** — define intent create/update/reconcile contract between `Document Engine` and `Revenue & Ops` (idempotency, status normalization, expiry/retry rules).
4. **Make Revenue & Ops consumers explicit** — cash flow and reports should pull through document-shaped APIs or views, not ad hoc duplicates (**supports Client Timeline + money story**).
5. **Publish payment UX contract in Experience checklist** — one status vocabulary + CTA matrix for document detail, public invoice, and portal paths.
6. **Stabilize Auth + Session** — execute **Foundation §4** (session/read/write matrix, invites, org bootstrap, documentation).
7. **Standardize UI layout** — execute **Foundation §5** + **A.2** table (`PageTemplate`, editor shell, embedded admin).
8. **Publish an "Experience checklist"** (1 page) for new screens touching documents or money — include the **Page Template** three-zone rule (`PageTemplate`) and **layout grammar** row picker (**closes Foundation §5**).
9. **Identity / role matrix doc** — admin vs management vs support capabilities (complements §4).
10. **Roll out** `PageTemplate` on Invoices, Quotes, Clients, and Services — move filters/summary into `sidePanel` where they still live inside the main card.
11. **Extend the hook → service → entity pattern** — add `*ListService` / query modules for Quotes, Clients, and other high-traffic reads; keep `api/InvoiceService`**-style** modules for non-CRUD delivery concerns.
12. **Implement Client Timeline** — query + UI on **Client detail** per **High impact §2**; consider a small `ClientTimelineService` for aggregation and caching keys.
13. **Implement quote → invoice conversion** — server or client path that creates draft invoice from accepted quote per **High impact §3**; validate RLS and idempotency (no duplicate drafts on double-accept).

Immediate execution order (exact)

1. **Add** `payment_intents` **table.**
2. **Integrate payment intents into the document lifecycle.**
3. **Expand** `document_events` **coverage and event ingestion.**
4. **Build payment UI into the sticky right panel on invoice/payment touchpoints.**
5. **Add basic reminders (event- and due-date driven).**

V1 IMPLEMENTATION CONTRACT (DIRECT BUILD PLAN)

Tables (Supabase)

- `payment_intents`: `id`, `org_id`, `source_kind` (`document` | `pos`), `document_id`, `pos_sale_event_id`, `provider` (`cash` | `ozow` | `card_terminal`), `amount`, `currency`, `status`, `external_id`, `idempotency_key`, `created_at`
- `document_events` (expand): ensure canonical events `sent`, `opened`, `clicked`, `paid`, `reminded` with `occurred_at`, `actor_type`, `metadata`
- `payments` (existing): invoice settlement/reconciliation only — not POS till money, not SaaS `payment_history`

Services (app/server orchestration)

- `PaymentIntentService` (`server/src/payments/`): create/confirm customer intents; POS checkout confirms through the provider registry. Idempotency by `org_id + idempotency_key`. PayFast is excluded.
- `DocumentEventService` (expand): append normalized document/payment lifecycle events; provide query helpers for timeline/reminders
- `PaymentWebhookReconciliationService` (new or expanded from current Payfast handler): verify provider payloads, map to canonical statuses, write `payments` + `document_events`, trigger settle transition
- `PaymentIntelligenceService` (v1 basic): evaluate reminder triggers (e.g., viewed-not-paid, due/overdue windows) and emit follow-up actions

Cron jobs (Vercel)

- `POST /api/cron/reminders` (existing pattern; expand logic): process due/overdue + behavior-based reminder candidates
- `POST /api/cron/payment-intelligence` (new): run lightweight trigger evaluation (`viewed>=N && not paid`, retry windows)
- `POST /api/cron/subscriptions-dunning` (existing/adjacent): keep tenant billing and subscription retries isolated from document settlement logic

API endpoints (`/api/*`)

- `POST /api/payment-intents` (create customer intent; POS or document)
- `GET /api/payment-intents/:id` (intent status)
- `GET /api/payment-intents/providers` (registered customer rails)
- `POST /api/payments/webhook/:provider` (customer rails, e.g. `ozow`; PayFast ITN stays on `/api/payfast-handler`)
- `POST /api/pos/webhook/:token` (POS sale ingress — generic, Yoco, Square parsers)
- `POST /api/pos/webhook/provider/square` (Square application webhook — routes by merchant_id)
- `POST /api/pos/oauth/square/start` · `GET /api/pos/oauth/callback/square` (Square OAuth connect)
- `POST /api/pos/oauth/yoco/connect` (Yoco API key connect + auto webhook)
- `GET|POST|PATCH|DELETE /api/pos/connections` (org settings managers / company admins); RLS mirrors this via `is_company_admin_for_org` (members may SELECT only)
- `GET /api/pos/sales` (`pos_access` for today; `pos_view_reports` for broader lists)
- `GET /api/pos/catalog` (`pos_access`; optional `register_id` — catalog is register-brand scoped)
- `POST /api/pos/checkout` (`pos_sell`; `pos_discount` if discount is applied)
- `POST /api/pos/return` (`pos_refund`; append-only return event; never deletes the original sale)
- `POST /api/pos/receipt/email` (till receipt email; not an invoice)
- `POST /api/pos/invoice` (optional tax-invoice copy of a completed sale; paid; no invoice payments)
- `GET|POST /api/pos/registers` · `PATCH|DELETE /api/pos/registers/:id` (till identity; org members list; settings managers write)
- `POST /api/documents/:type/:id/events` (new internal endpoint) or service-only ingestion path for `document_events`
- `POST /api/reminders/dispatch` (new internal endpoint used by cron workers)

v1 acceptance checks:

- Invoice can move `Deliver → Observe → Payment Intent → Settle` using provider-verified events only.
- Sticky right panel shows intent-aware CTA states (`Pay now`, `Retry payment`, `Send reminder`) without page-specific logic forks.
- `document_events` powers timeline entries and at least one automated reminder trigger.

End of document.